

ERP Replacement: Approach, Scope, and Data Strategy

TO Executive Team, Alpha Manufacturing Group
FRO Program Advisory
M

DATE 17 July 2026
RE Board-mandated ERP transformation

Bottom line

The age of the ERP isn't the problem. The problem is that we keep two systems of record for the same bill of materials, we can't cost a job while it's still open, and 80% of our custom orders lost margin last year because we found out after we'd already shipped. A cloud ERP is necessary. On its own it fixes none of that. What fixes it is one system of record for engineering and manufacturing data, labor captured against the job, and a reporting layer we own.

Recommendation: hold the 12-month date but redefine go-live. Wave 1 at month 12 puts every entity's finance and one operating site's full engineer-to-order cycle on the new platform. The remaining four sites follow in two waves over the next six months. We fit to standard, we migrate open items rather than history, and we build reporting outside the ERP so it survives the transition. Date is fixed. Scope is the variable.

1. What we're actually being measured on

- **Margin transparency.** Quoted versus actual contribution margin per job, visible while the job is open.
- **One BOM.** Engineering and manufacturing read the same record. The nightly sync gets retired, not improved.
- **No interruption.** We ship, invoice, and close every month throughout.
- **A simpler landscape.** Fewer systems, a named owner for every data domain, one set of KPI definitions.
- **An upgradable platform.** Standard configuration we can take quarterly releases on without a project.

What is deliberately *not* on that list: replacing every system, and reproducing the legacy ERP's behavior in a new tool.

2. Phasing: why site-by-site, and where we split

OPTION	ASSESSMENT
Big bang, five sites, month 12	One weekend, everything, everywhere. With 80% of orders custom, a BOM or MRP failure stops shipping at five plants at once. The blast radius is the company.
Module-phased, finance first	Keeps legacy alive as the MRP engine, so the fragile sync survives another year and the margin problem goes untouched for 18 months. We'd spend the budget on the part that isn't broken.
Site-phased, full core each wave (recommended)	All entity finance plus one site's operations at month 12; remaining sites in two waves. Proves the ETO process on real orders at limited blast radius, and each wave gets faster as the team learns. Cost: a temporary interface and roughly nine months of dual maintenance.

One deliberate split: all five entities move to the new GL together at Wave 1, so consolidation lives in one place from Day 1. Sites still on legacy feed summary journals until their wave arrives. We're splitting at the GL boundary rather than the entity boundary. That buys us a single close and costs us one interface. It's a trade worth making.

M0-M2	Mobilize. Sponsor and decision rights named. Data owners named. Item master and BOM cleanup starts in the legacy system. Auditors engaged on migration approach and revenue recognition treatment.
M2-M6	Fit-to-standard design. Conference room pilot run against five real historical orders, one of them with a mid-build engineering change. Scope freezes at the end of M6.
M6-M11	Build, test, three mock data loads, two mock cutovers. Mock cutover 2 includes a full trial month-end close on migrated data.
M12	Wave 1 go-live. All entity finance plus the pilot site. Timed to a fiscal quarter start. Not year end.
M13-M18	Waves 2 and 3. Legacy goes read-only. Decommission.
M18+	Value phase: CPQ, service management, advanced analytics.

3. Top risks

RISK	WHY IT'S REAL HERE	MITIGATION
Item master and BOM data quality	Twenty years of accumulation behind a sync that already fails.	Named owner per data domain. Cleanse in the legacy system, not in the load. Hard entry criteria before each mock. Every load reconciled and signed by Finance.
Scope creep and legacy habit	Eighty users with twenty years of muscle memory and a strong view on how it should look.	Fit-to-standard is the default. A change board that requires an executive sponsor and a quantified dollar impact per exception. Contingency held by steering, not by the PM.
SME capacity	Two hundred people. The ones who know the process are the ones running the plant.	Six named SMEs, funded, backfilled, ring-fenced from month 0. SME availability tracked weekly as a schedule risk, reported to steering.
Engineer-to-order process fit	Standard cloud ERP handles repetitive manufacturing well. Mid-build engineering change is where these programs die.	Prove it in the conference room pilot on real orders before we build anything. Make it a stage gate with the authority to stop the program.
Cutover stops shipments	We are a shipping business. A quiet dock is the failure mode that reaches the Board.	Shipment and receipt freeze window. Safety stock pre-built at the pilot site. Two rehearsed cutovers. A written rollback with a named decision owner and a fixed go/no-go hour.

A sixth, and we'll be blunt about it: nobody owns margin today. That's why it erodes. A system can't own a number. Assign a person.

4. Scope and sequencing: Day 1 versus deferred

IN AT WAVE 1 (NON-NEGOTIABLE)	DEFERRED BY DECISION, NOT BY DRIFT
GL, AP, AR, fixed assets, multi-entity, intercompany, consolidation, CAD/USD FX, US and Canadian tax.	CPQ and product configurator. Field service scheduling. Advanced planning and scheduling. WMS and barcoding. Supplier and customer portals. EDI expansion. Advanced analytics. Quality beyond basic nonconformance.
Item master, BOMs, routings, work centers, and a native ECN process . The engineering database retires here. This is the single biggest Day-1 win available to us.	CRM stays exactly where it is , untouched, and gets a light account and order interface in the value phase. HRIS and payroll stay siloed and post a summary journal. Neither is causing the margin problem and neither belongs in this cutover.
ETO order-to-cash end to end: quote, order, job, BOM, MRP, PO, production, ship, invoice.	Deferral isn't a downgrade. It's the mechanism that keeps Day-1 scope small enough to land safely. Everything here goes on a published Phase 2 backlog with a named owner, so it reads as sequenced rather than refused.
Procurement, receiving, inventory at cost.	
Shop floor time capture against job and operation. Without it there is no margin view, ever. It is the most likely thing to get cut and the least affordable.	
Job costing: estimate versus actual, live.	
Aftermarket parts and service order intake and billing. It's revenue. It doesn't get deferred.	
Revenue recognition exactly as we report it today, agreed with the auditors in M0-M2 rather than M11.	

Migration exclusions. We migrate active items, active BOMs and routings with activity in the last 36 months, active customers and vendors, on-hand inventory at cost, open POs, open sales orders, open jobs and WIP, open AR and AP, open ECNs, the opening trial balance, and two prior fiscal years of GL at period-summary level for comparatives. We do not migrate closed jobs, historical GL detail, paid invoices, superseded BOM revisions, items dormant for three years, or historical ECN detail. The rule we'll hold: **history is a reporting problem, not a migration problem.** We solve it with an archive, not with a load.

Customization discipline. Zero code at Day 1 is the target; configuration and reports only. An extension gets approved only if it clears three questions: is the gap proven in the pilot against a real order, is it customer-facing or regulator-facing, and who owns it in five years. Every custom object is built once and paid for at every quarterly release forever. Where we would otherwise customize, we change the process instead. That's the deal we sign when we go cloud, and it's cheaper to accept it now than in year three.

5. Data and reporting

Historical data. Two tiers. The ERP holds what it needs to transact; everything else goes to a read-only archive. We extract legacy to a cloud warehouse during design, keep legacy running read-only for six months after the final wave as insurance, then decommission it. Seven-year statutory retention is met by the archive. Nobody should be keeping a twenty-year-old ERP alive to satisfy an auditor.

Reporting through the transition. Build the reporting layer *outside* the ERP. A warehouse fed by both legacy and the new platform, with one conformed set of KPI definitions, is what actually delivers the uninterrupted access leadership is asking for. It also means the numbers don't quietly change on cutover weekend, which is the fastest way to lose executive trust in a new system.

- Agree and freeze roughly ten KPI definitions in M0-M2: revenue, gross and contribution margin by job, backlog, on-time delivery, quote-to-order conversion, WIP aging, inventory turns, labor efficiency, aftermarket attach.
- Through the phased window, group reporting comes from the warehouse mapped across both sources. Never from either ERP directly.
- Reporting change requests freeze from mock cutover 2 through Wave 1 plus 30 days.
- Expect a reporting dark period of about five business days at each cutover. We're saying so now rather than apologizing for it later.

Month-end close continuity. Wave 1 lands at a fiscal quarter start, not at year end and not into a close. We rehearse the close before we need it: mock cutover 2 includes a full trial close in the new system on migrated data, signed off by the Controller. The first two live closes will take longer, so we plan a three to five day extension, book the auditors and external accountants ahead of it, and run a close war room. Legacy stays open for one close cycle after Wave 1 for restatement, then goes read-only. The cutover itself is documented as a control event.

What we need from you

- 1 A single accountable sponsor with the authority to say no. CFO or COO. Not IT.
- 2 Six named SMEs, funded and backfilled, ring-fenced from month 0. This is the constraint that actually decides the date.
- 3 Agreement today that the date is fixed and scope is the variable. If both are fixed, quality becomes the variable, and quality failures at cutover stop shipments.
- 4 An owner for margin. Not a system. A person.